

# **AI Technologies in Smart Cities**

## **Industry Overview:**

The Healthcare Industry is no stranger to the use and implementation of AI models. Hospitals globally had used AI technology for numerous things, such as the detection of diseases, the assistance in monitoring and management of patients via virtual nurses, or aiding doctors and diagnosticians with difficult diagnoses. Even today, hospitals are still advancing in technology, including AI models.

## **Current AI Trends:**

Currently, there are many hospitals and healthcare companies around the world that are actively using AI models within their systems to aid in their work. For example, Memorial Hermann's "Hellocare" model assists in the monitoring of high-risk patients and patients in need of consistent movement. Hellocare also conducts virtual nurse meetings with patients, allowing for increased flexibility and a lesser workload for nurses and doctors alike. However, the need for face-to-face meetings with nurses is still something that is taken into account at these hospitals with virtual nurse integration.

Another hospital that implements AI into their systems is Houston Methodist. One of their AI technologies, “vICU (Virtual Intensive Care Unit)”, provides fast-paced and, in some cases, immediate care for ICU patients via virtual nurses, similar to Memorial Hermann’s Hellocare system. Another one of their AI systems, not publicly named, aids doctors in patient documentation and paperwork. This model is able to fill out documents relating to patients and their personal information. This does raise some questions about privacy and security if these systems were to be compromised.

Carenostics had come up with an ML model that predicts and identifies underdiagnosed cases of CKD (chronic kidney disease). The model aids doctors in the discovery of the diseases, highlights key symptoms that are the best indicators for CKD, which in turn lightens the burden of clinicians and doctors. Carenostics’ model has been deployed globally, with over 1000 models worldwide and 300 alone in the United States.

### Ethical Implications:

When it comes to ethics with AI in healthcare, there are several things that come up. Firstly, can these models be 100% trusted with the lives of humans? We know that the models can accurately diagnose patients and can highlight symptoms and irregularities within patients, but are they capable of saving a life if necessary?

If future advances in technology continue enough to the point where AI models begin to have a sense of sentience, is it possible that a model will make independent decisions based on its own experience? These are just some of the concerns that people have when it comes to AI in healthcare. Another obvious ethical concern to point out is the job displacement. With virtual nurses and AI diagnosticians, what is the point of having a human counterpart? Yes they are necessary for double checking and management of data, but once the models begin to show high-accuracies near the 100% accurate range, there will be little use for human medical professionals. The staff at these hospitals would drop significantly, with a majority of them being AI technicians managing the models, and some doctors/nurses for worse-case scenarios. This is an issue that is very relevant in other industries besides healthcare. But, there are still doubts that AI technology will completely take over the healthcare industry; you still simply need humans to care for other humans, since we understand ourselves best. It's not really an accuracy issue at this point, it's more of a preference quandary.

### Future Trends:

In the next few years or even decades, AI technology would advance far enough to where we would have minimal human interference with medical care. Virtual nurses would almost be a guaranteed feature, diagnoses and prescriptions

would be done completely by AI systems, and regular doctor visits would become less frequent for most people. Even now, we have the basic framework for AI models in healthcare. We have some virtual nurses, we have models who aid in the diagnosis of diseases and the recognition of symptoms, and we are seeing an overall decrease in face-to-face doctor visits. In the coming years, at least some form of AI technology would be implemented into hospitals around the world, with many different purposes in mind. However, with these new technologies comes new concerns, like the ones discussed in the ethics section. Job availability would probably be a priority, along with the trusting of AI with human lives at stake. We should still proceed cautiously with the rise of AI in many different industries, but we should also consider the positives of AI implementation.

### Reflection:

Personally, I'm on the fence when it comes to AI in our everyday industries. I do take into account the positives of having these technologies around: It makes our workloads lighter, it provides accurate information and results, and increases our reliability on AI as a whole. But, there are still negatives that I also consider, like job replacements, reliability to perform tasks, and security/privacy concerns if databases get breached. Overall, I think that AI in the future would be somewhat more stable and reliable. Honestly, I wouldn't mind working in the healthcare

industry, but not as a doctor or nurse. I would be the one managing the AI systems or using them to aid doctors in their own work. Database management would be a great start for this, since I've got some experience with AI models and databases. Only time will tell us about what will happen to AI in the future and how it will affect the economy.